Unit: 6 Unit 8: Quadratics
Topic: Solving by Quadratic Formula (Discriminant)

Name: ______________
Date: __________

Advanced (Habanero)

Q1. What is the discriminant of the quadratic equation $x^2 + 4x + 4 = 0$?
(A) 0
(B) 4
(C) 8
(D) 12
(E) 16

Q2. How many solutions does the equation $x^2+2x+1 = 0$ have?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4

Q3. Solve for x in $x^2-7x+12 = 0$ using the quadratic formula.
(A) $x = 3, x = 4$
(B) $x = 2, x = 5$
(C) $x = 1, x = 6$
(D) $x = 0, x = 7$
(E) $x = -1, x = 8$

Q4. What is the value of x in the equation $x^2 - 9x + 20 = 0$?
(A) $x = 4, x = 5$
(B) $x = 3, x = 6$
(C) $x = 2, x = 7$
(D) $x = 1, x = 8$
(E) $x = 0, x = 9$

Q5. The equation $x^2 + 2x + 5 = 0$ has
(A) two real solutions
(B) one real solution
(C) no real solutions
(D) infinitely many solutions
(E) undefined solutions

Q6. The discriminant of $ax^2 + bx + c = 0$ is $b^2 - 4ac$. What is the discriminant of $2x^2 + 5x + 3 = 0$?
(A) 13
(B) 17
(C) 25
(D) 29
(E) 37

Q7. Solve for x in $x^2 - 2x - 6 = 0$.
(A) $x = -2, x = 3$
(B) $x = 1pmsqrt7$
(C) $x = 2pmsqrt5$
(D) $x = -1pmsqrt8$
(E) $x = 0pmsqrt6$

Q8. For what value of k will the equation $x^2 + 4x + k = 0$ have one solution?
(A) $k = 2$
(B) $k = 4$
(C) $k = 6$
(D) $k = 8$
(E) $k = 10$

Open Ended Questions (FRQ)

Q9. Solve for x in $x^2 + 6x + 8 = 0$. Show all work and explain the number and type of solutions.

Q10. Find the discriminant of the quadratic equation $3x^2 + 2x - 5 = 0$. Determine the number and type of solutions based on the discriminant.

Chef's Tips

- Tip 1: Encourage students to check their work by plugging solutions back into the original equation.
- Tip 2: Remind students that the discriminant can be positive, zero, or negative, and that this determines the number and type of solutions.
- Tip 3: Emphasize the importance of showing all work and explaining the reasoning behind their answers, especially for free response questions.